

WHEN DOES POST-PRUNING FINE-TUNING HELP? TEMPORAL OPERATING-POINT SENSITIVITY IN TEXT-TO-AUDIO DIFFUSION EVALUATION

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ABSTRACT

Structured pruning of a text-to-audio diffusion model is usually followed by recovery fine-tuning, and the benefit of that fine-tuning is typically summarized by a single aggregate score at one inference setting. We show that, for the published pruned and post-fine-tuning checkpoints of AudioLDM-M-Full, the measured advantage of the post-fine-tuning checkpoint over the pruned checkpoint depends strongly on the temporal evaluation operating point. Using a paired, common-noise design with prompt-level bootstrap confidence intervals, we define a temporal interaction J as the difference between the fine-tuning advantage evaluated at the model’s native 10.24s scale and at a shorter 3.84s scale. A pre-registered domain-specialization hypothesis fails at the matched short operating point. The interaction J is positive but borderline at a milder pruning severity, and is prospectively confirmed at a stronger severity on a disjoint evaluation set ($J = +0.159$, 95% CI $[+0.131, +0.187]$), robust to a pruning-seam convention. The advantage is nonetheless still present at the short operating point, so it is not a 10s-only effect, and a severity-1 music-negative pattern does not replicate. An independent frame-level grounding evaluator corroborates a broad native-duration gain but rejects a simple late-event explanation. We conclude that recovery should not be treated as a checkpoint-intrinsic scalar and should be evaluated across inference operating points. We report explicit negatives and state that no matched dense fine-tuned control and no human evaluation are available, so the claims are about evaluation, not causal mechanism or perceptual quality.

Index Terms— text-to-audio generation, diffusion models, structured pruning, fine-tuning, evaluation, CLAP.

1. INTRODUCTION

Structured pruning followed by recovery fine-tuning is a standard route to cheaper generative diffusion models [3, 4], including for text-to-audio (TTA) generation with AudioLDM [1, 2]. The effect of the recovery stage is normally reported as the restoration of an aggregate benchmark number, such as Fréchet Audio Distance or a KL divergence on an in-domain test set at a fixed sampler setting.

A generative model, however, is queried at an inference operating point—clip duration, sampler steps, guidance—and there is no guarantee that a single aggregate number characterizes the fine-tuned checkpoint at every operating point a user might choose. In this paper we ask a narrow, empirical question: does the advantage of a post-pruning fine-

tuned checkpoint over the pruned checkpoint depend on the temporal evaluation operating point? We study this for the publicly released pruned and post-fine-tuning checkpoints of AudioLDM-M-Full, treating the released checkpoints as the object of study rather than proposing a new pruning or fine-tuning method.

Our contribution is an evaluation, not a mechanism. Concretely:

- A paired, common-noise evaluation with prompt-level bootstrap that isolates the post-fine-tuning—pruned contrast at two temporal operating points, and a temporal interaction J that quantifies how that contrast changes with duration.
- A prospectively frozen, cross-severity test: the interaction is borderline at a milder pruning severity and is confirmed at a stronger severity on a disjoint evaluation set with new seeds, robust to a pruning-seam convention.
- Honest negatives—a pre-registered domain-specialization hypothesis fails, the advantage at the short point does not vanish, a severity-1 music-negative pattern does not replicate, and an independent diagnostic rejects a simple late-event explanation—together with the single limitation (no matched dense fine-tuned control) that bounds the claim to evaluation.

2. RELATED WORK

Diffusion pruning and recovery. Structural pruning of diffusion U-Nets with fine-tuning recovery has been studied for image models [3, 4] and, for TTA, by Singh et al. [2], who prune an AudioLDM-family U-Net and recover with fine-tuning on the AudioCaps training set, reporting in-domain FAD/KL and event-capture metrics at a fixed sampler. These works evaluate recovery at a single operating point; we instead vary the operating point and measure the change in the fine-tuning advantage. Compression and inference regime. Compression error is known to interact with the reverse diffusion trajectory and the number of sampling steps. We are not aware of prior work establishing, for structural pruning of a TTA model, an evaluation-time interaction between the recovery advantage and the temporal operating point. TTA evaluation. CLAP similarity [5] is a standard automatic proxy for text–audio alignment; distributional metrics such as FAD [7] and event classifiers such as PANNs [6] are common but sample-size and encoder sensitive. These limits motivate a paired, multi-metric treatment.

3. METHOD

3.1. Systems and pruning severities

The dense model is AudioLDM-M-Full [1], a latent-diffusion U-Net with `channel_mult` = (1, 2, 3, 5) and ≈ 416 M U-Net parameters, itself AudioCaps- specialized by the original authors. We use the publicly released L1 structurally pruned checkpoints at two severities, denoted by the pruned per-stage channel multipliers: a milder severity 1 (1, 2, 3, 1) (145.7 M U-Net parameters, -65.0%) and a stronger severity 2 (1, 2, 1, 1) ($\approx 83\%$ U-Net reduction). Each severity prunes the deeper U-Net stages more aggressively. We verified that our pruning operator reproduces the released pruned weights bit-exactly (per-tensor oracle checks). We do not train any system.

3.2. Post-pruning fine-tuning checkpoints

The post-fine-tuning checkpoint at each severity is the released checkpoint obtained by fine-tuning the pruned U-Net on AudioCaps for 10^6 steps [2]; we load it and materialize its exponential-moving-average (EMA) weights, and verify its checksum. We refer to the pruned checkpoint and the post-fine-tuning checkpoint throughout; the recovery fine-tuning is AudioCaps fine-tuning of the pruned U-Net.

3.3. Evaluation batteries and operating points

The in-domain battery is AudioCaps test captions; the out-of-domain battery is a held-out music set of hip-hop-style captions derived from a MusicCaps-style source [9], with near-zero exposure to that subdomain in the recovery corpus. We generate at two temporal operating points: a short point (3.84 s, latent length 96) and the fine-tuning-native native point (10.24 s, latent length 256). All generation uses DDIM with 50 steps, guidance 2.5, $\eta = 0$, fp32, single generation, EMA weights. The music battery is generated only at the short point (its recipe).

The severity-2 evaluation uses disjoint sets constructed after the severity-1 results were known: 192 AudioCaps prompts that exclude the severity-1 prompts, the training set and the severity-1 music set, and 64 new music prompts with zero overlap with the severity-1 music set, all under new selection salts and generation seeds. Severity-1 uses 96 AudioCaps prompts for the controlled reversal test, a matched 80-prompt subset for the temporal comparison, and 64 music prompts.

3.4. Paired design, scoring, and statistics

For each prompt and operating point, the compared systems receive identical generation noise x_T (common random numbers), so a per-prompt contrast is not confounded by sampling noise. The primary scorer is CLAP (laion/clap-htsat-fused, revision 365dea6e) [5], applied with a fixed batch, order and seed convention that is part of the endpoint definition. The statistical unit is the prompt; the three music replicates are averaged to a per-prompt contrast before analysis. Confidence intervals are 95% prompt-level percentile bootstrap ($B = 10^4$).

3.5. Estimands and gates

We first ask, at a single operating point, how the fine-tuned checkpoint compares to the pruned one. Let $R(\text{context})$ be the mean per-prompt CLAP cosine of the post-fine-tuning checkpoint minus the pruned checkpoint in that context; positive values favour the post-fine-tuning checkpoint. We write R_{nat} , R_{short} and R_{music} for the AudioCaps native, AudioCaps short and held-out music contrasts.

The central quantity is whether the fine-tuning advantage changes with temporal scale. We define the temporal interaction

$$J = R_{\text{nat}} - R_{\text{short}}, \quad (1)$$

the difference between the advantage at the native and short operating points, estimated as a paired per-prompt contrast. A pre-specified gate declares the interaction resolved when the lower 95% bound satisfies $\log_5(J) > 0$. We also report a context contrast $K = R_{\text{nat}} - R_{\text{music}}$; because it changes both the prompt domain and the duration at once, we treat K as a bundled context effect, not a pure-domain effect. Sign-of-effect gates use a smallest effect size of interest SESOI = 0.025; a short-operating-point equivalence test asks whether the 90% CI of R_{short} lies within $[-0.025, +0.025]$.

3.6. Evidence classes

We separate: (i) prospectively frozen primary results (the severity-2 replication and its seam sensitivity; the pre-registered severity-1 negative), whose protocol and gates were frozen before scoring; (ii) a prospectively specified follow-up (the severity-1 temporal comparison); (iii) secondary corroboration (a second CLAP-family scorer, and KL/event-count metrics named in the frozen protocol but implemented after the primary result); and (iv) a post-result diagnostic (frame-level grounding). Cross-severity magnitude comparisons are exploratory and confounded (different experiments, prompts, sample sizes and checkpoints) and are never promoted to confirmatory evidence.

4. RESULTS

4.1. A controlled test rejects domain specialization

A severity-1 music observation motivated a strict hypothesis: that fine-tuning trades in-domain for out-of-domain alignment (a recovery reversal). We tested it prospectively at the matched short operating point (3.84 s). It fails: the in-domain contrast is essentially zero, $R_{\text{AC}} = -0.002$ ($n = 96$; Table 1), so there is no in-domain sacrifice to trade, and the pre-registered gate does not pass. This reframed the work from a mechanism claim to an evaluation question. The held-out music contrast at this severity is negative, $R_{\text{music}} = -0.094$.

4.2. A temporal operating-point hypothesis

The recovery fine-tuning operates at a native temporal scale of 10.24 s (latent length 256), whereas the controlled test above used 3.84 s (latent length 96). If the apparent absence of an advantage is specific to the short operating point, then evaluating at the native scale should change the picture. We therefore compare the fine-tuning advantage at the two temporal operating points directly.

Table 1. Primary CLAP contrasts (post-fine-tuning – pruned) with 95% bootstrap CI. Positive favours the post-fine-tuning checkpoint. †: CI excludes 0. ‡: pre-registered gate passed ($\log_5 > 0$).

Quantity	Sev. 1 (1, 2, 3, 1)	Sev. 2 (1, 2, 1, 1)
R_{nat} (10.24 s)	+0.052 [+0.009, +0.093] [†]	+0.244 [+0.215, +0.273] [†]
R_{short} (3.84 s)	+0.008 [-0.023, +0.039]	+0.085 [+0.066, +0.105] [†]
$J = R_{\text{nat}} - R_{\text{short}}$	+0.044 [-0.001, +0.087]	+0.159 [+0.131, +0.187] [†]
R_{music} (3.84 s)	-0.094 [-0.124, -0.065] [†]	+0.009 [-0.013, +0.032]
$K = R_{\text{nat}} - R_{\text{music}}$	—	+0.235 [+0.197, +0.272] [†]

n : sev. 1 AudioCaps 80 (matched), music 64; sev. 2 AudioCaps 192, music 64. Pre-registered sev. 1 reversal test (matched 3.84 s, $n=96$): $R_{\text{AC}} = -0.002$ [-0.027, +0.021], gate FALSE. K mixes domain and duration.

4.3. Severity 1: a larger native advantage, borderline interaction

On a matched 80-prompt subset, changing only the temporal extent yields a near-zero short advantage, $R_{\text{short}} = +0.008$, and a resolved native advantage, $R_{\text{nat}} = +0.052$ (Table 1). The interaction is positive but borderline, $J = +0.044$ [-0.001, +0.087]: the 95% CI narrowly includes zero, so the gate does not pass, and we report it as directional evidence (Fig. 1a). A second CLAP-family scorer gives a resolved interaction in the same direction ($J = +0.075$), but it is a corroborative secondary, not the primary endpoint.

4.4. Severity 2: a prospective replication

We then tested the pattern prospectively at the stronger severity 2 on the disjoint evaluation sets, with the estimands and gates frozen before any severity-2 output was scored. The interaction is clearly positive and resolved, $J = +0.159$ [+0.131, +0.187], driven by a large native-duration advantage, $R_{\text{nat}} = +0.244$ (Fig. 1b, Table 1). The context contrast is also resolved, $K = +0.235$. Both J and K are robust to the pruning-seam convention: an alternative baseline that reproduces the released checkpoint’s decoder-seam convention (differing in exactly three tensors) gives $J = +0.161$ and $K = +0.238$, with the same gate outcomes. The primary severity-2 baseline is the convention consistent with severity 1; the seam alternative is a pre-specified sensitivity analysis and never rescues a failed primary gate.

4.5. The advantage is present, not absent, at the short point

The interaction does not mean “fine-tuning helps only at 10s.” At severity 2 the short advantage is itself resolved and positive, $R_{\text{short}} = +0.085$, and the equivalence test fails (90% CI [+0.069, +0.102], outside the ± 0.025 margin): the advantage is smaller at the short point, not gone. The specific severity-1 music-negative pattern also does not replicate: at severity 2 the music contrast is approximately null, $R_{\text{music}} = +0.009$ (Table 1). Thus the context and temporal dependence replicate, but the exact severity-1 sign pattern does not. The stronger effects at severity 2 are consistent with a severity dependence, but a cross-severity comparison mixes two experiments, prompt sets and checkpoints and is treated as exploratory only.

Table 2. Severity-2 native-duration corroboration (post-fine-tuning vs. pruned). CLAP is primary; all others are secondary/corroborative and do not gate.

Metric	Value	Inference class
KL to ref. (↓)	$\Delta = +2.22$ [+1.93, +2.53]	paired CI
PANN capture count (↑)	$\Delta = +0.86$ [+0.70, +1.02]	paired CI
FD (PANN-2048, ↓)	48.7 vs. 107.2	descriptive
FAD (VGGish, ↓)	6.92 vs. 27.4	descriptive

4.6. Corroboration

We use secondary metrics conservatively; CLAP is the primary endpoint. At the severity-2 native point, a second CLAP-family scorer agrees on a large advantage ($R_{\text{nat}} = +0.375$, $J = +0.185$; both CIs exclude 0) but is not human evaluation; on held-out music this scorer shows a small negative contrast ($R_{\text{music}} = -0.037$) that the primary CLAP does not, with no gate role, so the music sign pattern remains unestablished. Two pre-specified event-level metrics, implemented after the primary result, also favour the post-fine-tuning checkpoint with paired CIs excluding zero (Table 2): a lower KL to real references and a higher count of PANN top-10 captured ground-truth labels. Distribution-level FAD and FD point in the same direction but are single scalars per system without paired CIs, so we report them as descriptive.

4.7. Frame-level diagnostic

As a prospectively frozen post-result diagnostic, an independent frame-level evaluator (a non-CLAP audio-text grounding model) measures where in the clip the requested events are grounded. At the native scale the post-fine-tuning checkpoint shows a large increase in frame-level grounding of requested events across the whole clip (semantic mass +0.27 [+0.22, +0.33], occupancy +0.27, quarter-coverage +0.38, peak +0.41; CIs exclude 0; both severities; seam-robust). But the gain is not concentrated late: the late-minus-early difference is essentially zero, $T = -0.002$ [-0.024, +0.020] at severity 2 (gate fails; -0.020 at severity 1, point not positive). The diagnostic therefore supports a broad native-duration grounding gain and rejects a simple explanation in which the benefit comes from events being shifted into the late portion of the clip. It is frame-level grounding evidence, not calibrated event probability, local causal activation, or perceptual quality.

5. DISCUSSION

The advantage of post-pruning fine-tuning over the pruned checkpoint is not a single number: in our automatic evaluation it depends on the temporal operating point and the evaluation context, and it is heterogeneous across prompts (some favour the pruned checkpoint on both CLAP and the frame-level diagnostic). The methodological implication is that recovery should not be treated as a checkpoint-intrinsic scalar: recovery claims for a generative model should be evaluated across several inference operating points rather than at one aggregate setting. We state this as an implication for evaluation, not a universal law—our study covers one model family, two pruning severities and one temporal axis.

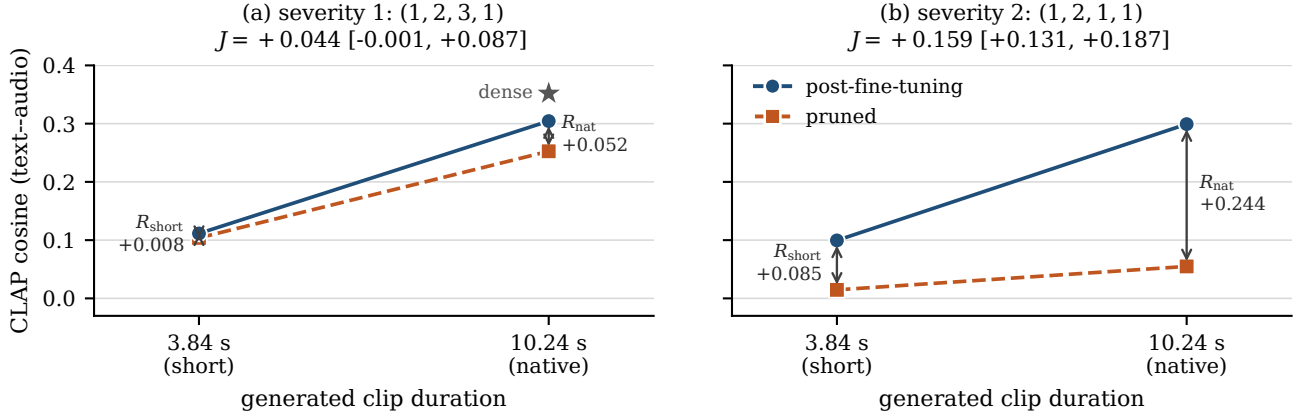


Fig. 1. The fine-tuning advantage grows with the temporal operating point. Mean CLAP cosine of the pruned and post-fine-tuning checkpoints at the short (3.84s) and native (10.24s) points, for (a) severity 1 and (b) severity 2. The vertical gaps are the paired contrasts R_{short} and R_{nat} ; the temporal interaction $J = R_{\text{nat}} - R_{\text{short}}$ (with 95% CI) is annotated per panel. The advantage is present at both points but larger at the native scale, and the interaction is borderline at severity 1 and resolved at severity 2. The dense reference (severity 1, native) is a secondary control. Note the y -axis is a mean score, so the interaction is the change in the gap, not the overall rise.

6. LIMITATIONS

The main limitation is that mechanistic attribution is blocked. The ideal control—a dense, unpruned model given a matched AudioCaps fine-tune—is unavailable (the checkpoint was confirmed deleted), and the public dense text-fine-tuned release is not an equivalent substitute (different data and recipe, dense start). We therefore cannot attribute the observed dependence to pruning specifically, and we do not claim that fine-tuning restores the pruned model, that it restores dense performance, or that it consistently improves individual samples; the dense reference (Fig. 1a) neither resolves a residual gap to dense nor establishes equivalence to it. The primary scorer family is CLAP (the second scorer is CLAP-derived), and there is no human evaluation: a designed listening study was cancelled before launch with zero participants, so it contributes no evidence, and the public audio examples are illustrative only. The operating point differs from the published recipe (DDIM 50 vs. 200 steps), so external validity to that recipe is untested although both systems are scored identically. The cross-severity comparison is exploratory and confounded, and the frame-level diagnostic is a post-result grounding measure, native-duration only.

7. CONCLUSION

For the released pruned and post-fine-tuning checkpoints of AudioLDM-M-Full, the advantage of fine-tuning over pruning depends on the temporal evaluation operating point: an interaction that is borderline at a milder pruning severity is prospectively confirmed at a stronger severity on a disjoint set, robust to a pruning-seam convention, present at the short point, and corroborated by an independent frame-level measure. Post-pruning recovery is better described as multi-operating-point behaviour than as a single restored score.

Companion audio examples are online.¹

8. REFERENCES

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¹<https://gbibbo.github.io/audioldm-modality-swap-pruning/>